

Txy Guided Inquiry: Benzene-Toluene VLE

Chemical Engineering Thermodynamics

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Abstract

This guided inquiry uses an interactive Temperature-composition (T -xy) diagram to explore Vapor-Liquid Equilibrium (VLE), boiling behavior, Antoine vapor pressures, and the Lever Rule for binary mixtures. Use this document alongside the interactive **Benzene-Toluene Txy Explorer**.

1 Introduction to the T -xy Diagram

A T -xy diagram displays the phase behavior of a binary mixture at a constant, specified system pressure. The boundaries define the regions of subcooled liquid, superheated vapor, and the two-phase vapor-liquid equilibrium (VLE) region.

1.1 Key Boundaries and Regions

- **Bubble-Point Curve (x_1):** The boundary representing the temperature at which the first bubble of vapor forms upon heating a liquid of known composition. Below this line, the system is entirely a **subcooled liquid**.
- **Dew-Point Curve (y_1):** The boundary representing the temperature at which the first drop of liquid condenses from a vapor upon cooling. Above this line, the system is entirely a **superheated vapor**.
- **Vapor-Liquid Equilibrium (VLE) Region:** The two-phase region enclosed between the bubble-point and dew-point curves. Within this region, liquid and vapor phases coexist in dynamic equilibrium.

Active Exploration Task 1:

Open the **Benzene-Toluene Txy Explorer**. Ensure the system pressure is set to 760 mmHg.

1. Click in the area at 85°C and $z_{\text{benzene}} = 0.3$. Note the reported phase.
2. Click in the area at 105°C and $z_{\text{benzene}} = 0.7$. Note the reported phase.
3. Click in the area at 95°C and $z_{\text{benzene}} = 0.5$. Note the reported phase.

2 Pure Component Boiling Points

The boiling point of a pure substance at a given pressure corresponds to the temperature where its vapor and liquid phases are in equilibrium. On a T -xy diagram, this corresponds to the two horizontal extremes ($x_1 = 0$ and $x_1 = 1$).

2.1 Boiling Points of Benzene and Toluene

By looking at the extremes of the diagram:

- At $x_{\text{benzene}} = 0$, the system contains 100% Toluene (Component 2).
- At $x_{\text{benzene}} = 1$, the system contains 100% Benzene (Component 1).

Active Exploration Task 2:

With the pressure set to 760 mmHg:

1. Click on the far-left boundary ($x = 0$) and record the temperature. This is the normal boiling point of pure Toluene.
2. Click on the far-right boundary ($x = 1$) and record the temperature. This is the normal boiling point of pure Benzene.
3. Based on these temperatures, which component is more volatile? Explain.

Boiling Points Discussion ($P = 760$ mmHg):

Pure Benzene ($x_{\text{benzene}} = 1.0$) has a normal boiling point of approximately 80.1°C .

Pure Toluene ($x_{\text{benzene}} = 0.0$) has a normal boiling point of approximately 110.6°C .

Benzene has a lower boiling point, which means it has a higher vapor pressure at any given temperature. Therefore, **Benzene is the more volatile component.**

3 The VLE Region & Tie-Lines

When the overall mixture composition (z_1) and temperature (T) fall inside the two-phase envelope, the system splits into a liquid phase of composition x_1 and a vapor phase of composition y_1 . These equilibrium compositions are connected by a horizontal line called a **tie-line**.

Active Exploration Task 3:

Click inside the VLE envelope at 100°C and $z_{\text{benzene}} = 0.4$.

1. Locate the blue dot on the bubble line. Write down its coordinate (x_{benzene}).
2. Locate the red dot on the dew line. Write down its coordinate (y_{benzene}).
3. Compare y_{benzene} and x_{benzene} to the overall composition $z_{\text{benzene}} = 0.4$. What does this tell you about which phase is enriched in the more volatile component?

Tie-Line Concept Verification:

At 100°C and 760 mmHg, the equilibrium compositions are:

$$x_{\text{benzene}} \approx 0.256 \quad \text{and} \quad y_{\text{benzene}} \approx 0.459$$

Notice that $y_{\text{benzene}} > z_{\text{benzene}} > x_{\text{benzene}}$. The vapor phase is significantly enriched in Ben-

zene (the more volatile component), while the liquid phase is stripped of Benzene and enriched in Toluene.

4 The Lever Rule and Phase Amounts

While the endpoints of a tie-line define the *compositions* of the coexisting phases, the relative *amounts* of the liquid and vapor phases are calculated using a material balance known as the **Lever Rule**.

Let n be the total moles of system, n_L be the moles of liquid, and n_V be the moles of vapor. The total mole balance and component mole balances yield:

$$n = n_L + n_V$$

$$z_1 n = x_1 n_L + y_1 n_V$$

Solving these equations simultaneously gives the fraction of the system in the liquid phase (L) and vapor phase (V):

$$L = \frac{n_L}{n} = \frac{y_1 - z_1}{y_1 - x_1} \quad (1)$$

$$V = \frac{n_V}{n} = \frac{z_1 - x_1}{y_1 - x_1} \quad (2)$$

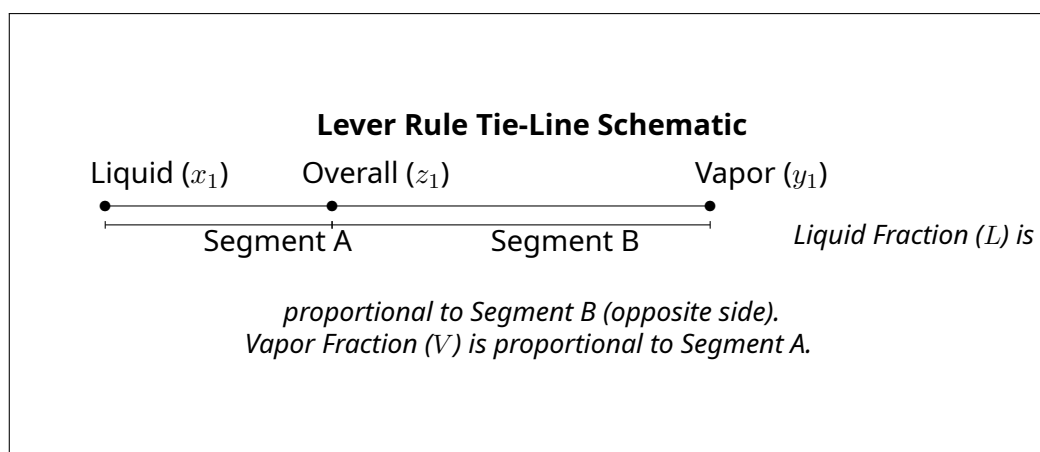


Figure 1: Visual mechanical representation of the Lever Rule along a horizontal tie-line.

Worksheet Problem:

Suppose you feed a binary mixture containing 40 mol% Benzene ($z_{\text{benzene}} = 0.40$) and 60 mol% Toluene to a flash drum operating at 98°C and 760 mmHg.

1. Click as close as possible to $z_{\text{benzene}} = 0.40$ and $T = 98^\circ\text{C}$ in the simulator.
2. Record the equilibrium liquid and vapor compositions (x_1, y_1).
3. Use Equations (1) and (2) to manually calculate the mole fraction of the system that remains as liquid (L) and the fraction that flashes to vapor (V). Show your calculations.
4. Compare your hand-calculated results with the "Phase Amounts" bar in the Explorer.

Problem Solutions & Calculations:

At 98°C and 760 mmHg:

- $x_{\text{benzene}} \approx 0.283$
- $y_{\text{benzene}} \approx 0.495$

Using the Lever Rule equations:

$$L = \frac{y_1 - z_1}{y_1 - x_1} = \frac{0.495 - 0.400}{0.495 - 0.283} = \frac{0.095}{0.212} \approx 0.448 \quad (44.8\% \text{ Liquid})$$

$$V = \frac{z_1 - x_1}{y_1 - x_1} = \frac{0.400 - 0.283}{0.495 - 0.283} = \frac{0.117}{0.212} \approx 0.552 \quad (55.2\% \text{ Vapor})$$

These hand-calculated percentages match the visual indicator reported by the simulator interface.

5 The Effect of Pressure on Boiling Behavior

Phase equilibria are highly dependent on system pressure. Under elevated pressures, the molecules in a liquid phase require more kinetic energy (and thus higher temperatures) to overcome intermolecular forces and transition to the vapor state.

Active Exploration Task 5:

Observe how the T -xy diagram responds to changing pressure constraints:

1. Note the temperature values on the Y-axis at the default 760 mmHg.
2. Navigate to the **Constant Pressure** input field. Change the pressure from 760 mmHg to 2000 **mmHg** and hit Enter.
3. Observe the visual changes on the chart. What happened to the bubble and dew curves?
4. Find the new boiling point of pure Benzene ($x_{\text{benzene}} = 1.0$) at this elevated pressure.

Pressure Shift Discussion:

When the pressure is increased from 760 mmHg to 2000 mmHg:

- The entire two-phase envelope **shifts upward** along the temperature axis.
- At 2000 mmHg, pure Benzene boils at approximately 120.4°C, compared to its normal boiling point of 80.1°C at 760 mmHg.
- Pure Toluene's boiling point shifts from 110.6°C to approximately 155.1°C.
- This demonstrates that the system temperature must be significantly higher to achieve vapor-liquid coexistence under higher pressures, which aligns with Antoine's equation predictions.